Lecture Tutorial 2A: Mathematical tools for Physics II

Snapshot of what mathematical concepts are required for Physics II:

- vector fields & flux
- line & surface integrals
- gradients, divergence & curl

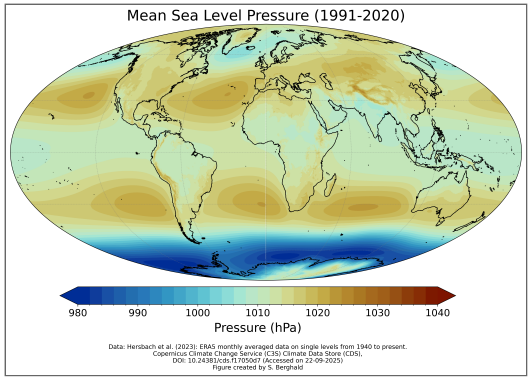
Reference for this tutorial and self-studying:
https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)/16%

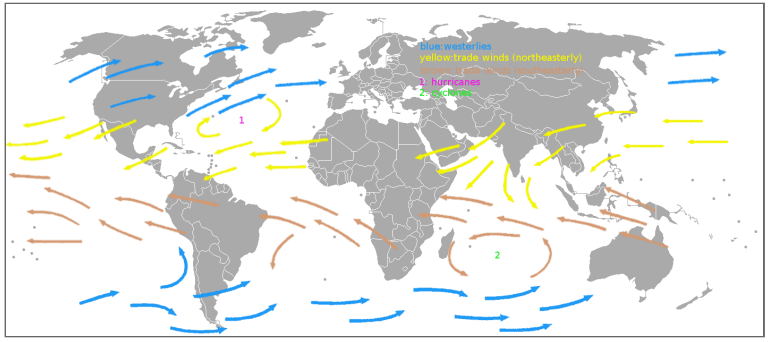
Fields instead of particles

- Physics I: study physics by observing individual particles (positions, velocities, forces)
- Physics II: study physics by assigning quantities to every point $\Rightarrow$ fields

from ***wikipedia***, ***CC0 1.0 Universal***

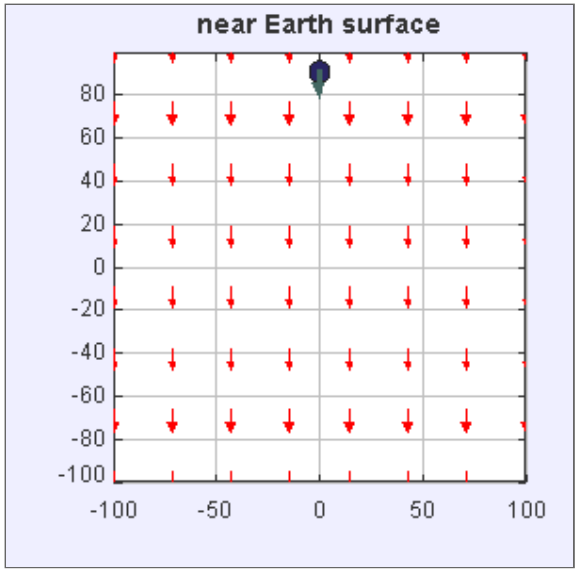
Example of scalar and vector field

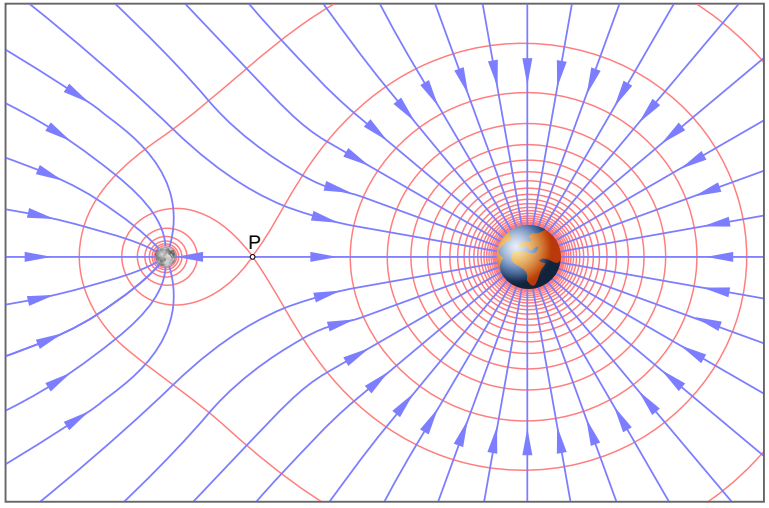




*[left] from **wikipedia**, public domain; [right] from **wikipedia**, **Attribution-Share Alike 4.0 International***

Gravitational field: g is not a scalar





*[left] from **wikipedia**, **Attribution-Share Alike 3.0 Unported**; [right] from **wikipedia**, **Attribution-Share Alike 4.0 International***

From vectors to vector fields (2D example)

Vector: Magnitude and direction, visualized as arrows

$$\vec{v} = v_x \hat{i} + v_y \hat{j} \qquad \text{or} \qquad \vec{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

Vector field: Assigns a vector to every point in space

$$\vec{F}(x, y) = -y\,\hat{i} + x\,\hat{j} \qquad \text{or} \qquad \vec{F}(x, y) = \begin{bmatrix} -y \\ x \end{bmatrix}$$

$\Rightarrow$ **At each point (x, y), the field assigns a vector whose components depend on position.**

Example:

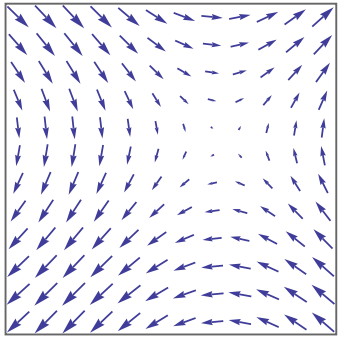
At $(1,0)$: $\vec{F} = (0,1)$

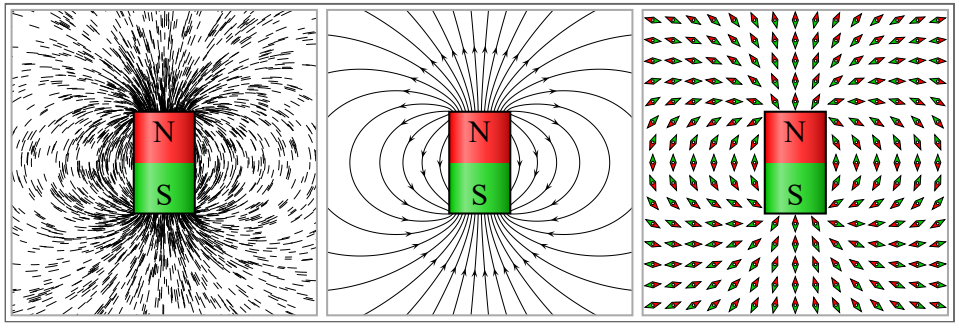
At $(0,1)$: $\vec{F} = (-1,0)$

Vector fields: How to visualize

Two complementary representations help us understand a vector field:

- **Arrows (local view):** vectors at *selected points* show magnitude and direction
- **Field lines (global view):** *continuous* curves reveal the overall structure and flow





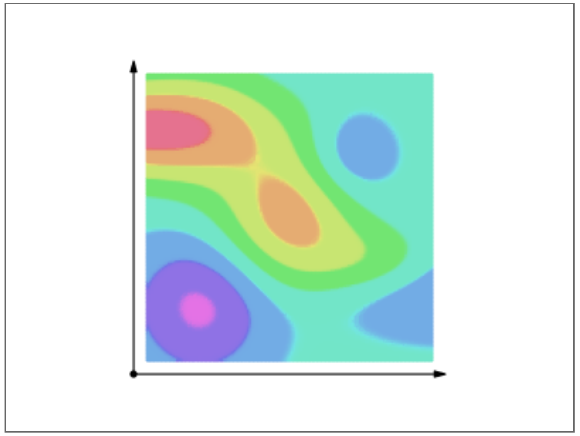
*[left] from **wikipedia**, public domain; [right] from **wikipedia**, **Attribution-Share Alike 4.0 International***

Line integrals of scalar fields

A scalar line integral accumulates a scalar quantity along a path:

$$\int_C f\,ds$$

- Idea:
  - Split path C into small elements ds
  - ds measures arc length along the curve
  - Sum values $f\,ds$ along the path
- No direction dependence, only magnitude matters, but f can vary along the path and may be positive or negative
- Result is a single scalar quantity representing the total amout along the path
- Scalar line integrals generalize $\int f(x)\,dx$ to curves in space



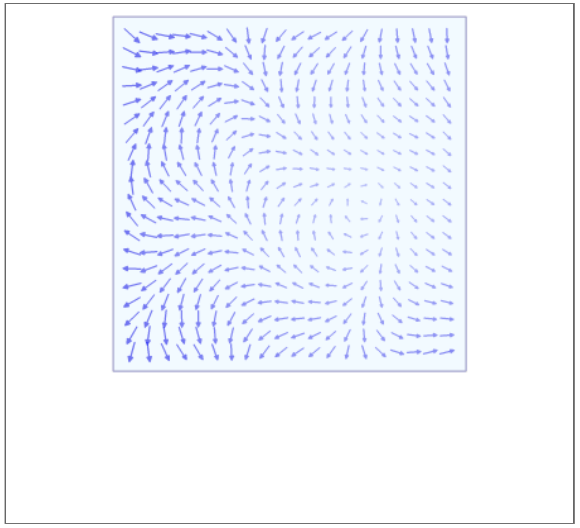
*from **wikipedia**, public domain*

Line integrals of vector fields

A vector line integral measures how a field acts along a path:

$$\int_C \vec{F} \cdot d\vec{l}$$

- Extends scalar line integrals by considering direction of path and vector field
- Idea:
    - Split path C into small segments $d\vec{l}$
    - $d\vec{l}$ points along the path
    - Sum contributions $\vec{F} \cdot d\vec{l}$ along the path
- Direction matters through the dot product, only the component along the path contributes
- Result is a single scalar measuring the total effect along the path



*from **wikipedia**, public domain*

Closed line integrals & conservative fields

- **Closed line integral** follows a path that starts and ends at the same point:

$$\oint \vec{F} \cdot d\vec{l}$$

- For **conservative fields**:

$$\Rightarrow \oint \vec{F} \cdot d\vec{l} = 0$$

- For a closed loop, start and end coincide
- **Intuition:** moving "forward" and coming back cancels exactly $\Rightarrow$ no net effect remains
- $\Rightarrow$ **Path independence** for conservative fields

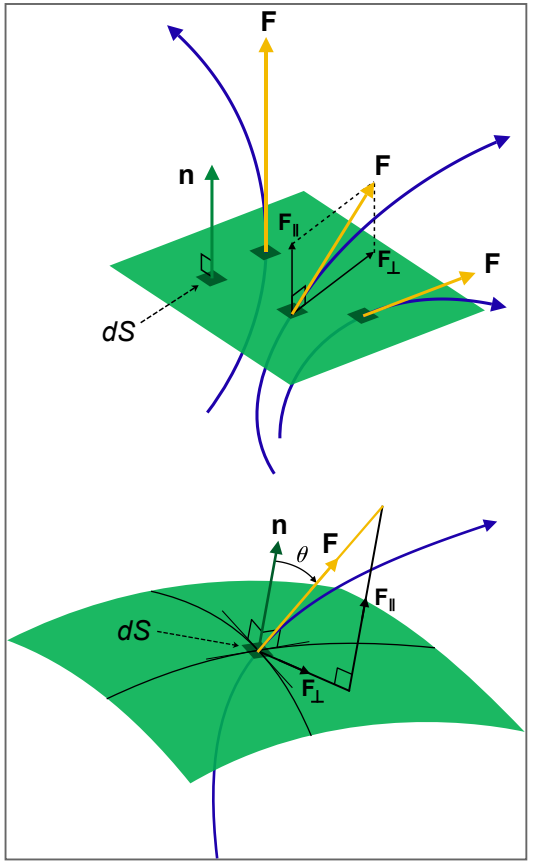
- If a closed integral is **non-zero** $\Rightarrow$ the field has circulation (not conservative)

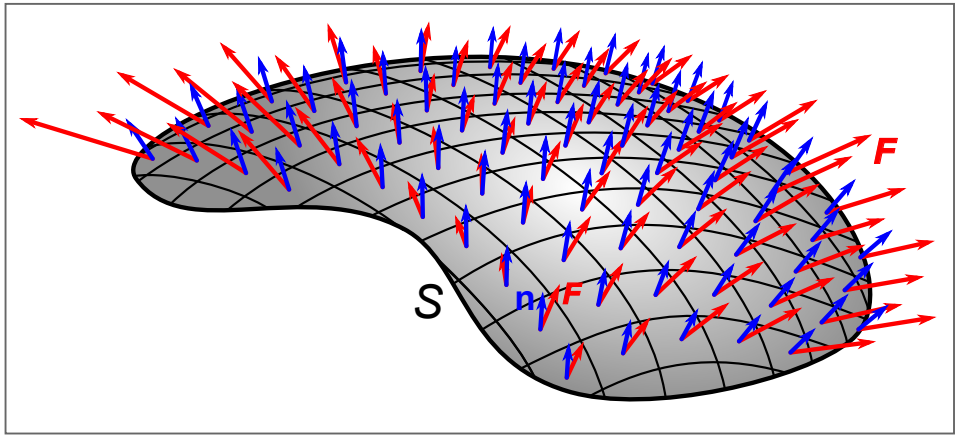
Flux & surface integrals: How much field passes through a surface

Flux Φ measures how much of a vector field $\vec{F}$ flows through a surface S:

$$\Phi = \int_S \vec{F} \cdot d\vec{A}$$

- Flux is the surface integral of a vector field
- **Idea:**
  - Split surface S into small patches $d\vec{A} = \hat{n}\, dA$ (normal $\hat{n}$)
  - Dot product $\Rightarrow$ only component through surface contributes
  - Sum contributions $\vec{F} \cdot d\vec{A}$ over the surface
- Field aligned with normal $\rightarrow$ maximum
- Field parallel to surface $\rightarrow$ zero
- Result is a single scalar: total flow through the surface





[left:] from ***wikipedia***, ***CC0 1.0 Universal***; *[right:] from* ***wikipedia***, ***CC0 1.0 Universal***

Gradient: From scalar field to direction of change

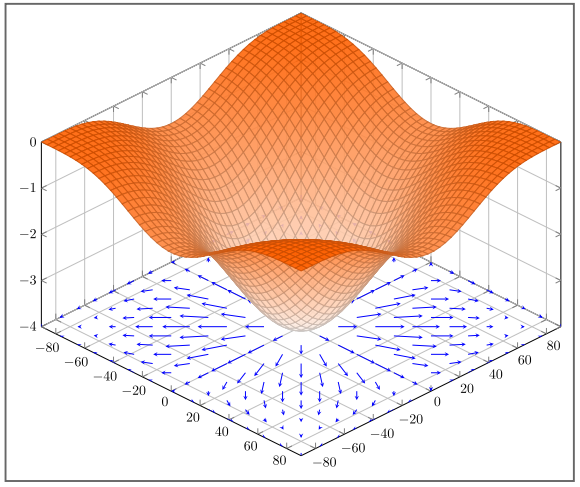
The **gradient** ∇V points in the direction of strongest increase of $V(x, y, z)$, thus converts a scalar field into a vector field:

$$\nabla V$$

$$\nabla V = \begin{bmatrix} \frac{\partial V}{\partial x} \\ \frac{\partial V}{\partial y} \\ \frac{\partial V}{\partial z} \end{bmatrix} \qquad \text{or} \qquad \nabla V = \frac{\partial V}{\partial x}\,\hat{i} + \frac{\partial V}{\partial y}\,\hat{j} + \frac{\partial V}{\partial z}\,\hat{k}$$

- **Idea:**
  - Measure how V changes in each spatial direction, i.e. take the derivative
  - Combine spatial derivatives into a vector $\rightarrow$ **for each point, we know the direction of strongest increase**

- Result is a vector field describing change in space
- ∇V is perpendicular to surfaces of constant V
- Magnitude: $|\nabla V|$ gives strength of change



from ***wikipedia***, ***CC0 1.0 Universal***

Example: Gradient of a simple scalar field

Given a scalar field:

$$V(x, y) = x^2 + y^2$$

Compute the gradient:

$$\nabla V = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

- Gradient points radially outward
- Magnitude increases with distance from origin
- Direction = steepest increase of V

From scalar to vector field derivatives

We can take derivatives of scalar fields:

$$\nabla V(x, y, z) \quad (\text{gradient})$$

What about vector fields $\vec{F}(x, y, z)$?

- A vector field has multiple components
  $\Rightarrow$ derivatives can be combined in different ways
- Apply ∇ to a vector field:
  - Dot product $\Rightarrow$ divergence $\nabla \cdot \vec{F}$
  - Cross product $\Rightarrow$ curl $\nabla \times \vec{F}$
- These operations extract different physical information
  - Divergence: measures sources and sinks
  - Curl: measures rotation of the field

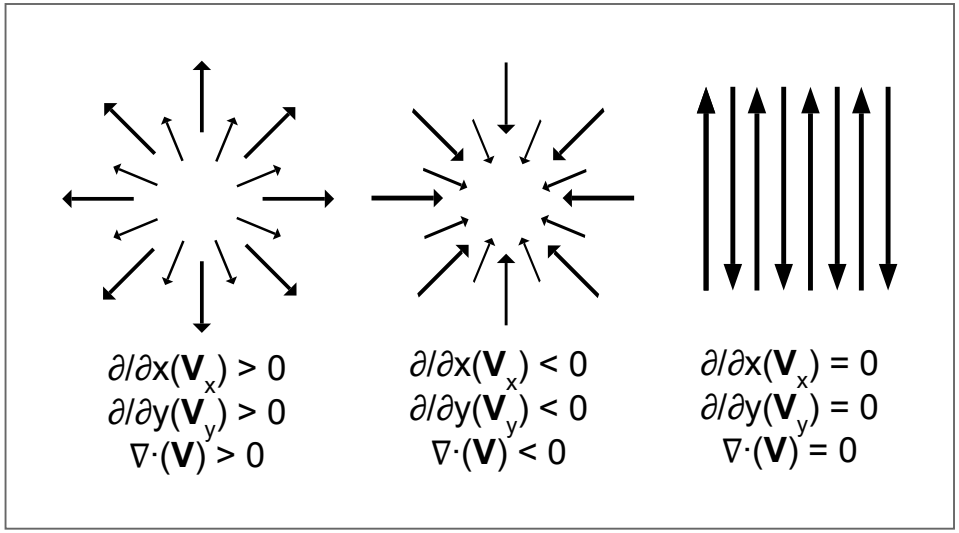
Divergence: Sources and sinks of a field

Divergence measures how a vector field spreads out or converges:

$$\nabla \cdot \vec{F}$$

$\Rightarrow$ **Differentiate each component along its own direction and sum the results**

- **Conceptual idea:**
  - Consider a small volume around a point
  - Compare inflow and outflow through the surface
  - Assign net outflow to the point
  - Positive $\rightarrow$ source (flow outward)
  - Negative $\rightarrow$ sink (flow inward)
  - Zero $\rightarrow$ no net source or sink
- **Result is a scalar field describing local source strength**



from ***wikipedia****.svg),* ***Attribution-Share Alike 4.0 International***

Example: Divergence of a vector field

Given a vector field:

$$\vec{F}(x, y) = \begin{bmatrix} x^2 \\ 2y + 1 \end{bmatrix}$$

Compute the divergence:

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} = \frac{\partial x^2}{\partial x} + \frac{\partial (2y + 1)}{\partial y} = 2x + 2$$

- Divergence depends on position
- Positive for $x > -1 \rightarrow$ source
- Negative for $x < -1 \rightarrow$ sink

- Results is a **scalar field** (single number which can depend on position but does not have direction/components)

Curl: Local rotation of a field

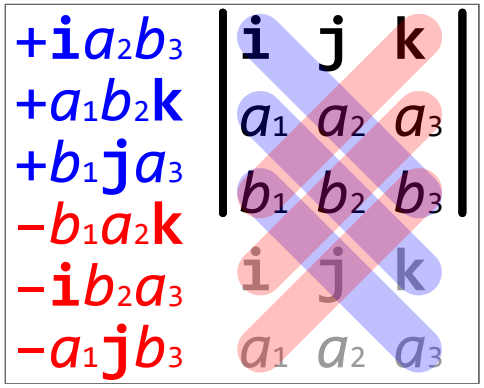
Curl measures how a vector field $\vec{F}(x, y, z)$ rotates or circulates:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ F_x & F_y & F_z \end{vmatrix} = \begin{bmatrix} \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{bmatrix}$$

$\Rightarrow$ **Combine derivatives of components across directions via a cross product**

- **Conceptual idea:**
  - Consider a tiny paddle wheel at a point
  - Check whether the field causes rotation
  - Direction $\rightarrow$ axis of rotation (right-hand rule)
  - Magnitude $\rightarrow$ strength of rotation
  - Zero $\rightarrow$ no local rotation

- **Result is a vector field describing local circulation**



from ***wikipedia***, ***Attribution-Share Alike 4.0 International***

Example 1: Curl of a vector field

Given vector field from divergence example:

$$\vec{F}(x,y) = [\begin{matrix} x^2 \\ 2y+1 \end{matrix}] \quad \Rightarrow \quad \vec{F}(x,y,z) = \begin{bmatrix} x^2 \\ 2y+1 \\ 0 \end{bmatrix}$$

Note: The curl points perpendicular to the plane of local rotation, not necessarily perpendicular to the vector field itself

Curl is defined as a cross product with ∇:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ x^2 & 2y+1 & 0 \end{vmatrix} = \begin{bmatrix} \frac{\partial 0}{\partial y} - \frac{\partial(2y+1)}{\partial z} \\ \frac{\partial x^2}{\partial z} - \frac{\partial 0}{\partial x} \\ \frac{\partial(2y+1)}{\partial x} - \frac{\partial x^2}{\partial y} \end{bmatrix}$$

$$= \hat{k}\,(\frac{\partial(2y+1)}{\partial x} - \frac{\partial x^2}{\partial y}) = \hat{k}(0-0) = 0$$

- Curl is zero $\rightarrow$ no local rotation
- Divergence $\neq 0 \rightarrow$ sources/sinks present
- $\Rightarrow$ Spreading $\neq$ rotating

Example 2: Curl of a vector field

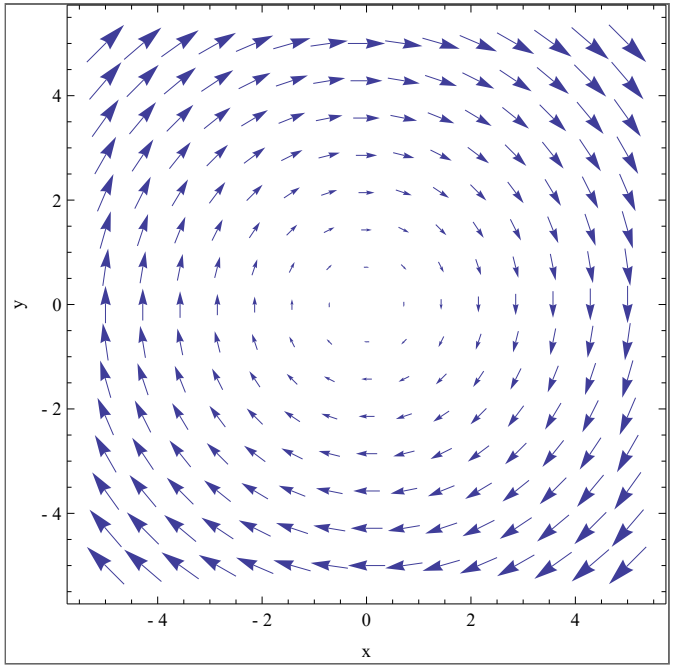
Given vector field:

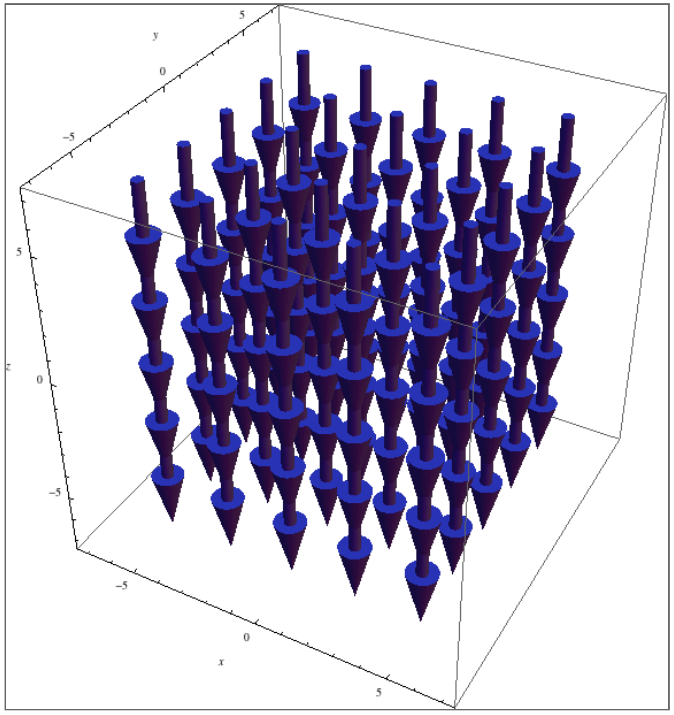
$$\vec{F}(x, y, z) = \begin{bmatrix} y \\ -x \\ 0 \end{bmatrix}$$

Curl is defined as a cross product with ∇:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ y & -x & 0 \end{vmatrix} = \begin{bmatrix} \frac{\partial 0}{\partial y} - \frac{\partial(-x)}{\partial z} \\ \frac{\partial y}{\partial z} - \frac{\partial 0}{\partial x} \\ \frac{\partial(-x)}{\partial x} - \frac{\partial y}{\partial y} \end{bmatrix} = \hat{k}\,(\frac{\partial(-x)}{\partial x} - \frac{\partial y}{\partial y}) = \hat{k}(-1-1) = -2\hat{k}$$

- Result is a **vector field** (here along $-\hat{k}$)
- Curl is constant and non-zero
- Field rotates around the origin (clockwise) & field forms closed loops


4
2
0
- 2
- 4
y
- 4
- 2
0
2
4
x


y
5
0
−5
z
5
0
−5
x
−5
0
5

*[left] from **wikipedia**, **Attribution-Share Alike 3.0 Unported**; [right] from **wikipedia**, public domain*

Example 3: Curl of a vector field

Given vector field:

$$\vec{F}(x, y, z) = \begin{bmatrix} 0 \\ -x^2 \\ 0 \end{bmatrix}$$

Curl is defined as a cross product with ∇:

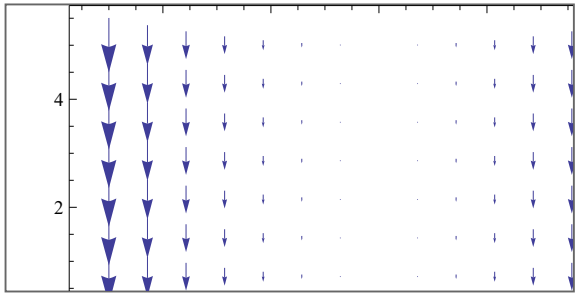
$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ 0 & -x^2 & 0 \end{vmatrix} = \begin{bmatrix} \frac{\partial 0}{\partial y} - \frac{\partial(-x^2)}{\partial z} \\ \frac{\partial 0}{\partial z} - \frac{\partial 0}{\partial x} \\ \frac{\partial(-x^2)}{\partial x} - \frac{\partial 0}{\partial y} \end{bmatrix} = \hat{k}\,(\frac{\partial(-x^2)}{\partial x} - \frac{\partial 0}{\partial y}) = \hat{k}(-2x - 0) =$$

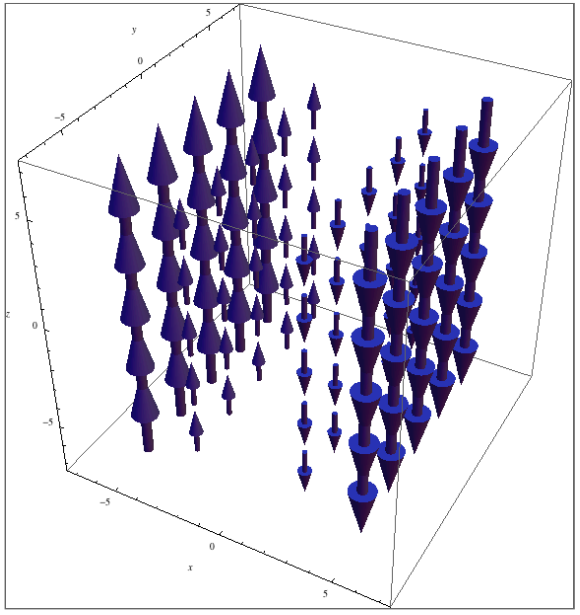
$$-2x\,\hat{k}$$

- Curl depends on position
- Zero at $x = 0$, increases with $|x|$

- Direction changes sign with x
- Shear flow: velocity varies with position

  $\Rightarrow$ neighboring layers move differently and induce local rotation
- Shear flow: Flow speed varies with position $\Rightarrow$ neighboring layers slide past each other $\Rightarrow$ a small paddle wheel would start rotating





*[left] from **wikipedia**, public domain; [right] from **wikipedia**, public domain*

Versatility of the Del / nabla operator

$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} \qquad \text{or} \qquad \nabla = \hat{i}\,\frac{\partial}{\partial x} + \hat{j}\,\frac{\partial}{\partial y} + \hat{k}\,\frac{\partial}{\partial z}$$

Gradient:

$$\mathrm{grad}\, f = \nabla f$$

Divergence:

$$\mathrm{div}\, v = \nabla \cdot v$$

Curl:

$$\operatorname{curl} \mathrm{v} = \nabla \times \mathrm{v}$$

Flux and sources (Gauss' theorem)

The total flux through a closed surface S equals the sum of sources inside the surface:$\oint_S \vec{F} \cdot d\vec{A} = \int_V (\nabla \cdot \vec{F}), dV$ $

- Left: total flux through the surface S
- Right: divergence integrated over the volume V
- **Conceptual idea:**
    - Divergence measures local sources (per volume)
    - Integrating over a volume gives total source strength
    - Equals total flux through the boundary surface
    - $\Rightarrow$ Flux = accumulated divergence

Circulation and rotation (Stokes' theorem)

The circulation along a closed path equals the total rotation inside:

$$\oint_{\partial S} \vec{F} \cdot d\vec{r} = \int_S (\nabla \times \vec{F}) \cdot d\vec{A}$$

- Left: circulation along the boundary ∂S
- Right: curl integrated over the entire surface S
- **Conceptual idea:**
  - Curl measures local rotation (circulation density)
  - Integrating curl over a surface gives total circulation
  - local (curl) → integrate → global (circulation)
  - $\Rightarrow$ Circulation = accumulated curl

Why is the curl of a conservative field zero?

- In conservative fields, any closed path has zero circulation:

$$\oint_{\partial S} \vec{F} \cdot d\vec{l} = 0$$

- **Using Stokes' theorem:**

$$\oint_{\partial S} \vec{F} \cdot d\vec{l} = \int_S (\nabla \times \vec{F}) \cdot d\vec{A}$$

$$\Rightarrow \int_S (\nabla \times \vec{F}) \cdot d\vec{A} = 0$$

- This must hold for any surface S

  $\Rightarrow$ the integrand must vanish

$$\nabla \times \vec{F} = 0$$

- **Result: conservative fields are irrotational (no local rotation)**

Why is the curl of a gradient zero?

$$\nabla \times (\nabla V) = \begin{bmatrix} \frac{\partial}{\partial y}(\frac{\partial V}{\partial z}) - \frac{\partial}{\partial z}(\frac{\partial V}{\partial y}) \\ \\ \frac{\partial}{\partial z}(\frac{\partial V}{\partial x}) - \frac{\partial}{\partial x}(\frac{\partial V}{\partial z}) \\ \\ \frac{\partial}{\partial x}(\frac{\partial V}{\partial y}) - \frac{\partial}{\partial y}(\frac{\partial V}{\partial x}) \end{bmatrix} = \begin{bmatrix} \underbrace{\frac{\partial^2 V}{\partial y \partial z} - \frac{\partial^2 V}{\partial z \partial y}}_{=0} \\ \\ \underbrace{\frac{\partial^2 V}{\partial z \partial x} - \frac{\partial^2 V}{\partial x \partial z}}_{=0} \\ \\ \underbrace{\frac{\partial^2 V}{\partial x \partial y} - \frac{\partial^2 V}{\partial y \partial x}}_{=0} \end{bmatrix} = \vec{0}$$

- Order of differentiation does not matter, $\partial_i \partial_j V = \partial_j \partial_i V$
- $\Rightarrow$ all terms cancel pairwise
- $\Rightarrow$ no local rotation
- **Result:** $\nabla \times (\nabla V) = 0$ **(gradient fields are irrotational)**

Why is the divergence of a curl zero?

$$\nabla \cdot (\nabla \times \vec{F}) = \frac{\partial}{\partial x}\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \frac{\partial}{\partial z}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

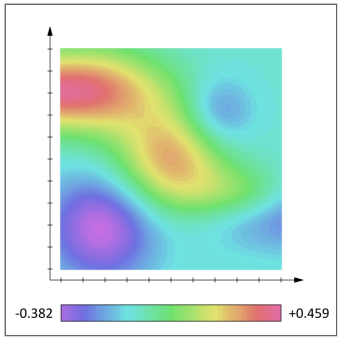
$$= \underbrace{\frac{\partial^2 F_z}{\partial x \partial y} - \frac{\partial^2 F_z}{\partial y \partial x}}_{=0} + \underbrace{\frac{\partial^2 F_x}{\partial y \partial z} - \frac{\partial^2 F_x}{\partial z \partial y}}_{=0} + \underbrace{\frac{\partial^2 F_y}{\partial z \partial x} -}_{=0}$$

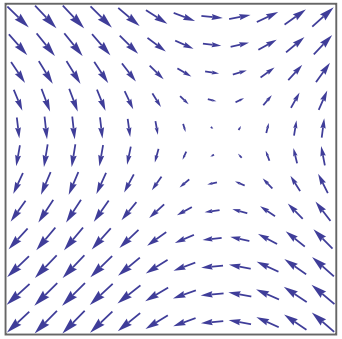
- Curl describes local circulation
- Divergence measures sources or sinks
- Consider a small volume:

  $\Rightarrow$ inflow and outflow balance exactly

  $\Rightarrow$ **Rotation alone cannot create a net outflow**

Revisit: Mathematical tools for Physics II





*[left] from **wikipedia**, public domain; [right] from **wikipedia**, public domain*

- **Inside the coffee (bulk fluid):** incompressible flow
  $\Rightarrow$ what flows in = what flows out
  $\Rightarrow \nabla \cdot \vec{v} = 0$
- **At the hole (leak):** fluid leaves the cup (sink)
  $\Rightarrow$ more leaves than enters
  $\Rightarrow \nabla \cdot \vec{v} < 0$